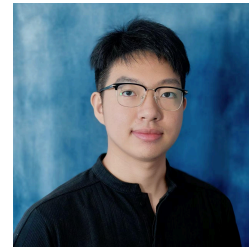


ZHONGYANG FAN

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EDUCATION

University of California, Los Angeles (UCLA)

Master of Applied Statistics and Data Science Expected Dec 2026

GPA: 4.0/4.0

Relevant Coursework: Deep Learning, Large Language Models, Data Management, Applied Regression, Mathematical Statistics

University of California, Davis

B.S. in Applied Mathematics Dec 2024

GPA: 3.79/4.0

Honors: Three-times Dean's Honors List

Relevant Coursework: Numerical Analysis, Linear Algebra, Stochastic Processes, Real Analysis, Complex Analysis

EXPERIENCE

Vetology

Data Science Intern | Jan 2026-Mar2026

- Created a gold-standard labeled dataset by quality-checking 300 canine and feline thoracic and abdominal imaging reports for an automated veterinary report scoring system.
- Applied prompt engineering to develop and refine structured prompts for ChatGPT and Gemini across multiple scoring criteria, and built batch API workflows for large-scale testing and side-by-side model comparison.
- Measured model performance with confusion matrices, sensitivity, and specificity, and improved scoring consistency by iterating on prompts and scripts based on team feedback and experiment results.
- Achieved over 90% sensitivity and about 98% specificity with the best model and prompt setup.

Energy Innovation Capital

Data Analyst Extern | Sep 2024-Dec 2024

- Researched the U.S. and global electric vehicle charging market, analyzing macro trends, industry challenges, risks, and emerging companies, and summarized findings in investment-focused reports.
- Cleaned and analyzed company data in Python and Excel, and built an evaluation framework using metrics such as headcount, funding rounds, and total assets to assess startup growth potential.
- Assessed promising energy startups, including Spirii, using venture capital and industry research methods; findings supported internal investment discussions and business planning.

PROJECTS

E-commerce User Purchase Prediction from Clickstream Data

Team Project | Aug 2023-Oct 2023

- Cleaned YooChoose clickstream and purchased data and engineered features such as dwell time, click count, number of viewed products, and holiday windows; applied rebalancing strategies to address class imbalance.
- Helped build and compare Random Forest, GBDT, and XGBoost models using 4-10 day sliding windows for purchase prediction; the best model achieved 79.9% accuracy.
- Developed RNN and LSTM models and improved performance with Dropout and L2 regularization, achieving 0.82 AUC and approximately 76.8% accuracy.
- Compared machine learning and deep learning approaches on accuracy and generalization to support model selection for e-commerce purchase intent prediction.

LEADERSHIP & ACTIVITIES

Surreal Number Seminar, Department of Mathematics

Team Member | Sep 2024-Dec 2024

- Organized and led seminar discussions on surreal number construction, proof strategies, and combinatorial models.
- Proposed an alternative framework for defining surreal numbers, formalized rules for addition, subtraction, and multiplication, and verified algebraic consistency and closure.
- Applied surreal number theory to Hackenbush game analysis to identify optimal strategy paths.

SKILLS

Programming: Python, R, MySQL

Tools: Excel, Microsoft Office, LibreOffice

Languages: Mandarin Chinese (native), English (professional working proficiency)